2c. Content of Principles of Design and Technology (J310/01)

The specification content is set out through an enquiry approach to support teaching and learning. In addition, 'core' and 'in-depth' principles are highlighted throughout the content to demonstrate the required levels of knowledge, understanding and application that learners should develop in relation to each part of the content.

The '**core**' principles of Design and Technology offer a broad set of principles that all learners must know regardless of their specific practical experiences. These principles are required so that learners are able to make informed choices as a designer and demonstrate their fundamental knowledge and understanding of the subject. The 'core' content of the exam draws and builds on prior knowledge covered in the Key Stage 3 curriculum of study.

Where learners are required to demonstrate their '**in-depth**' knowledge, understanding and design development skills, this will relate to the specific experiences they have pursued throughout their Design and Technology learning. When designing and/or making, learners will build an in-depth toolkit of knowledge, understanding and skills in relation to materials or systems they have worked with and have an interest in, this can be drawn upon to support written responses.

Learners should build in-depth knowledge, understanding and design development skills that relate to one or more of the following main categories of materials:

- papers and boards
- natural and manufactured timber
- ferrous and non-ferrous metals
- thermo and thermosetting polymers
- natural, synthetic, blended and mixed fibres, and woven, non-woven and knitted textiles.

Alternatively some learners may choose to follow the design engineering requirements if they have more of an interest and in-depth understanding of electronic and mechanical systems and control. This path still requires much of the same in-depth knowledge, but there will be more of a focus on parts 6.3 and 6.4 of the content.

In the written examination, all learners are required to demonstrate their mathematical skills and scientific knowledge as applied to design and technology practice. The level of mathematical and scientific knowledge within this qualification should be equivalent to Key Stage 3 learning.

The mathematical requirements and possible applications are outlined at the end of this section and in Appendix 5c. Where mathematical skills are learnt through 'in-depth' areas, learners should be able to apply this learning to other broader contexts. The scientific knowledge is integrated into the content and outlined in Appendix 5d.

Symbols are used to clearly identify examples where mathematics and/or science could be considered relevant:

 = Maths

 = Science

The subject content should be underpinned by understanding and applying it to a range of contextual approaches that allow learners to develop their skills, knowledge and understanding through iterative designing, innovation and communication; studying materials and technologies; working with materials and technologies; considering manufacture and production methods; critiquing; exploring existing products and considering values and ethics.

The content has two columns to indicate with a tick (✔) whether the content relates to 'core' principles or 'in-depth' principles to support appropriate levels of teaching and learning.

Where content is listed using a Roman numeral bullet e.g. (i), it denotes content that **must** be taught and may be directly assessed in the examination. Where content is listed using bullet points '•' or '○' or follows an e.g., this content is illustrative only and does not constitute an exhaustive list. A direct question will not be asked about the examples listed but learners will need to draw on such examples when responding to questions in the examination.